

Comparison with Kernel-Based, Texture-Based, and Small-Data Cancer Classification Studies

Work	Cancer / modality	Dataset scale reported	Representation	Classifier / model	Reported result	Comparison to our project
Doyle et al. [1]	Breast histopathology	48 breast biopsy tissue studies	Gabor texture plus architectural features; spectral clustering for dimensionality reduction	SVM	95.8% cancer vs. non-cancer accuracy with Gabor texture features	Closest small-data breast histopathology precedent. Our work uses a broader analytic kernel bank, patient-level patch splits, MMR/QMC selection, heatmaps, and held-out patch AUC/accuracy.
Ojansivu et al. [2]	Breast cancer morphology in histopathology	1,092 TMA images	LBP and LPQ texture histograms	RBF-kernel SVM with chi-square distance metric	62.0% three-class accuracy; 0.95 AUC and 90% accuracy for extreme class 1 vs. 3	Uses classic texture descriptors and SVM. Our task is binary tumor/background patch classification from annotation masks, with explicit kernel families and selected filter responses.
Gopinath and Shanthi [3]	Thyroid FNAC cytology	80 training images and 30 test images	Gabor-filter-bank statistical texture after ROI segmentation	SVM	96.7% accuracy, 95% sensitivity, 100% specificity	Strong example of a very small dataset and Gabor/SVM pipeline. Our method generalizes this idea by combining many analytic filter families and selecting compact discriminative subsets.
Zheng [4]	Breast mammography	DDSM subset	Circular Gaussian filtering for alarm generation plus Gabor features for false-alarm reduction	CAD classification pipeline	Validated by ROC/AUC on DDSM subset	Related through circular/Gabor filtering and breast cancer detection. Our method uses patch-level annotation-derived labels and reports kernel-bank/CNN held-out results.
Kadhim and Kamil [5]	Breast invasive ductal carcinoma histology	Random subsets of 100, 200, 400, 1000, and 2000 images	Gabor statistical features	Multiple classical classifiers including SVM	Best results reported at 100 images and wavenumber 0.2	Directly relevant to the requested 100-image regime. Our project uses more images/patches and studies interpretable kernel selection rather than one Gabor feature family.
He et al. [6]	Cervical histopathology	20 patients, 135 WSIs, 3,000 patches	First-order histogram, K-means clustering, GLCM, nucleus morphology	SVM	96.8% accuracy when combining four feature groups	Very close in spirit: patch extraction, small patient count, texture/morphological handcrafted features, and SVM. Our work differs by using convolutional kernel responses, MMR diversity selection, QMC sampling, and radial/composite kernel studies.
Cai et al. [7]	Renal cancer histopathology	Histopathology dataset; paper emphasizes small/medium dataset robustness	Deep AlexNet features fused with HoG, GLCM, LBP, Gabor, and MRF texture descriptors	Hybrid deep-plus-texture classifier	Fusion outperformed AlexNet alone and single texture descriptors	Highly relevant feature-family overlap: GLCM, LBP, Gabor, MRF. Our project avoids large deep feature dependence in the main interpretable model and directly selects analytic kernels.
Karuppasamy et al. [8]	Breast histopathology	Two small breast cancer datasets, including SQUH and BreaKHis experiments	Feed-forward filters learned outside back-propagation by logistic regression or SVM modules	CLR / CSVM-H	Faster training and stronger performance than VGGNet-16/ResNet-50 on SQUH in reported experiments	Closest modern small-data alternative to back-propagation CNNs. Our method is more explicitly analytic and interpretable because filter families and selected kernels are inspectable.

Work	Cancer / modality			Dataset ported	scale re-	Representation	Classifier / model	Reported result	Comparison to our project
Ours	Breast patches	medical	image	10,070 patches from local malignant/normal TIFF images and annotations	extracted from local	Multi-family analytic kernel responses; MMR selection; Sobol QMC sampling; radial, composite, and invariant variants	Logistic classifier / shallow MLP; CNN baseline for reference	Best selected-kernel model: 0.9387 test AUC and 0.8866 test accuracy; CNN baseline: 0.9626 test AUC and 0.9021 test accuracy	Combines classical interpretability with systematic kernel-bank generation, patient-level splitting, compact ablations, heatmap visualization, and comparison against a learned CNN baseline.

Short Interpretation

The most directly comparable small-data/kernel papers are Doyle et al., Gopinath and Shanthi, Kadhim and Kamil, and He et al. They show that fixed texture descriptors and SVMs remain competitive when image counts are limited. Cai et al. is especially relevant because it uses the same broad texture families as our implementation, including GLCM, LBP, Gabor, and MRF, but it fuses them with deep AlexNet features. Our contribution is different: we build a broad analytic convolutional kernel bank, score and select kernels by discrimination and response diversity, and evaluate compact interpretable response sets against a CNN baseline.

Strict Small-Data Kernel Screening

The following screening table applies the stricter criterion requested for this project: small data with at most about 200 original samples, a non-CNN classifier, and classification based on one to three trained or selected kernels. None of the reviewed prior papers satisfies all three criteria exactly.

Paper	Small data ≤ 200 ?	1–3 kernels?	Not CNN?	Match quality
Doyle et al. [1]	Yes, 48 biopsy studies	No, many Gabor/textural features	Yes, SVM	Small-data and non-CNN, but not 1–3 kernels
Gopinath and Shanthi [3]	Yes, 110 FNAC images total	Close, evaluates individual Gabor wavelength/angle settings	Yes, SVM	Closest match
Kadhim and Kamil [5]	Yes, includes 100-image subset	No, Gabor statistical feature extraction	Yes, classical ML including SVM	Small-data and non-CNN, but not explicitly 1–3 kernels
He et al. [6]	Original data: 20 patients / 135 WSIs, but 3,000 patches	No, four feature groups	Yes, SVM	Small patient count, but patch count > 200
Ojansivu et al. [2]	No, 1,092 images	No	Yes, SVM	Not small enough
Cai et al. [7]	Not clearly ≤ 200	No	Hybrid with AlexNet	Not a match
Karuppasamy et al. [8]	Small datasets, exact count not in accessible abstract	No	Not standard CNN backpropagation, but convolutional/feed-forward	Partial only

This makes the compact-kernel part of our project comparatively distinctive: the repository explicitly reports one-kernel, three-kernel, and five-kernel MMR-selected classifiers, e.g. one selected kernel reaching 0.9157 test AUC and three selected kernels reaching up to 0.9239 test AUC, while still comparing against larger kernel banks and a CNN baseline.

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